

Zero-Dimensional Measure Extraction in Compact Spaces

An audit, closure theorems, and an exact theorem for arbitrary products

Research note on Problem 7.3 of Głodkowski

18 June 2026

Abstract

Głodkowski asked for a description of the compact Hausdorff spaces K for which every nonzero Radon measure charges a compact zero-dimensional subset. This note audits a partial automatic solution and develops it substantially. The approximation property used in that solution is not stronger than the original positive-measure property: the two are equivalent. The extension theorem also has no exceptional-set hypothesis, because every extension in the source paper is a closed subspace of $K \times [0, 1]$. Both the original property and its compact G_δ version are preserved by countable products, countable inverse limits, and all countable towers of those extensions.

A ZFC nonexample follows from Plebanek’s compact connected space carrying a normal probability measure. A more elementary family of nonexamples is obtained from uncountable cubes. The main new result of this note gives an exact characterization for an arbitrary product $\prod_{i \in I} K_i$: besides requiring each factor to have the extraction property, only countably many factors may support a Radon measure whose support is not zero-dimensional. Consequently, for compact metrizable factors M_i , the product $\prod_i M_i$ has the property exactly when only countably many M_i are non-zero-dimensional. In particular, $[0, 1]^I$ has the property if and only if I is countable.

No complete classification of all compact Hausdorff spaces was found in the literature search. The problem is repeated, still as a problem, in Głodkowski’s 2024 doctoral thesis. Thus the results below are a corrected and substantially enlarged partial answer, not a claimed solution of the full classification problem.

1 The problem and the status of the literature

The question in [3, Problem 7.3] is:

Describe the class of compact Hausdorff spaces K such that for every nonzero Radon measure μ on K there is a zero-dimensional compact subset $L \subseteq K$ such that $\mu(L) \neq 0$.

The same question appears as Problem 3.5.3 in the author’s April 2024 doctoral thesis [4]. Searches by the exact problem text, title and arXiv identifier, together with searches using the phrases “zero-dimensional compact subset”, “Radon measure”, “normal measure”, “measure-separable compacta”, and related product/inverse-limit terminology, did not locate a subsequent complete classification through 18 June 2026.

There is relevant earlier literature. Zindulka proved that every finite measure on a metric space has a full-measure zero-dimensional subspace [8]. Džamonja and Kunen developed analogous closure methods for the different class of compacta on which every Radon measure

has separable measure algebra [1]. Plebanek constructed a compact connected space carrying a normal probability measure [6]; this supplies a ZFC nonmember of the class considered here. None of these sources gives the requested general classification.

Throughout, “zero-dimensional” means covering dimension zero. For compact Hausdorff spaces this is equivalent to having a base of clopen sets. All measures are finite Radon measures unless stated otherwise.

2 Audit of the partial result

Let the *positive extraction property* mean that every nonzero finite positive Radon measure on K charges a compact zero-dimensional subset. The partial packet called the following property “strong zero-dimensional measure extraction”: for every positive μ and every $\eta > 0$, there is compact zero-dimensional $Z \subseteq K$ with $\mu(K \setminus Z) < \eta$.

The central conclusions of the packet are sound, but four corrections materially strengthen and clarify them.

- (1) The approximation property is *equivalent* to positive extraction; it is not a genuinely stronger hypothesis. This is proved in Theorem 3.2.
- (2) The packet’s metric proof is a valid repair of a small gap in the proof of [3, Theorem 3.16]. An arbitrary decreasing local base need not have summable boundary tails, and the asserted eventual-tail family need not automatically be a base. Choosing balls with distinct, tail-dense radii fixes both points; see Theorem 4.2.
- (3) The exceptional-set assumption in the packet’s extension theorem is unnecessary. An extension L in the source paper is, by definition, a compact subspace of $K \times [0, 1]$. Heredity and product closure therefore give the result for *every* extension, strong or not; see Corollary 5.2.
- (4) The packet’s final discussion is too pessimistic about compact G_δ extraction. The compact G_δ version is also hereditary and closed under countable products and countable inverse limits, hence under every countable tower of source-paper extensions. This does not by itself remove metrizability from the source construction: Proposition 6.2 also uses weak-star sequential compactness of bounded sequences of measures, and later arguments use further metric input.

There is also a bibliographic slip: the packet refers to “Proposition 5.9” of the source paper, but Section 5 ends with Corollary 5.8. The relevant use of the metric extraction theorem and a decreasing neighborhood sequence occurs in Proposition 6.2.

3 Equivalent formulations

Definition 3.1. *A compact Hausdorff space K belongs to the class $\mathbf{E}$ if every nonzero finite positive Radon measure μ on K has a compact zero-dimensional set $Z \subseteq K$ with $\mu(Z) > 0$.*

We use the standard closed-sum theorem for covering dimension: if a normal space is a countable union of closed subspaces of dimension at most n , then it has dimension at most n [2, Theorem 3.1.8]. In particular, a finite union of compact zero-dimensional subspaces of a Hausdorff space is again compact and zero-dimensional.

Theorem 3.2 (Extraction, approximation, and decomposition). *For a compact Hausdorff space K , the following are equivalent.*

- (i) $K \in \mathbf{E}$.
- (ii) *For every finite positive Radon measure μ on K and every $\eta > 0$, there is a compact zero-dimensional $Z \subseteq K$ such that*

$$\mu(K \setminus Z) < \eta.$$

- (iii) *Every finite positive Radon measure μ on K can be written*

$$\mu = \sum_{n=1}^{\infty} \mu_n$$

as a sum of positive Radon measures such that $\text{supp } \mu_n$ is zero-dimensional for every n .

The sum in (iii) is an increasing sum of positive measures and converges in total variation.

Proof. Assume (i), and fix μ . Put

$$s = \sup\{\mu(Z) : Z \subseteq K \text{ is compact and zero-dimensional}\}.$$

Choose compact zero-dimensional Z_n with $\mu(Z_n) > s - 2^{-n}$, and replace Z_n by $Z_1 \cup \dots \cup Z_n$. Thus (Z_n) is increasing, every Z_n is compact zero-dimensional, and

$$\mu\left(\bigcup_n Z_n\right) = s.$$

If $s < \mu(K)$, define the positive Radon measure

$$\lambda(A) = \mu\left(A \setminus \bigcup_n Z_n\right), \quad A \in \text{Bor}(K).$$

Restriction of a finite Radon measure by a Borel set is again Radon. By (i), some compact zero-dimensional W satisfies $\lambda(W) > 0$. For all sufficiently large n ,

$$\mu(Z_n \cup W) \geq \mu(Z_n) + \lambda(W) > s,$$

a contradiction. Hence $s = \mu(K)$, proving (ii).

Assume (ii). Choose an increasing sequence of compact zero-dimensional sets Z_n with $\mu(K \setminus Z_n) < 2^{-n}$. Set $Z_0 = \emptyset$, $B_n = Z_n \setminus Z_{n-1}$, and

$$\mu_n(A) = \mu(A \cap B_n).$$

Then $\text{supp } \mu_n \subseteq Z_n$, so every support is zero-dimensional, and $\mu = \sum_n \mu_n$ because $\mu(K \setminus \bigcup_n Z_n) = 0$. This proves (iii).

Finally, if (iii) holds and $\mu \neq 0$, then some μ_n is nonzero. With $S_n = \text{supp } \mu_n$ we have

$$\mu(S_n) \geq \mu_n(S_n) = \mu_n(K) > 0,$$

and S_n is compact zero-dimensional. Thus (i) holds. □

Proposition 3.3 (Positive, signed, and complex measures). *Property E is equivalent to the formulation in Problem 7.3 for nonzero real signed Radon measures. It is also equivalent to the analogous formulation for complex Radon measures.*

Proof. Only the converse direction needs proof. Let ν be a nonzero real signed measure. Replacing ν by $-\nu$ if necessary, assume $\nu^+ \neq 0$. Choose a Hahn-positive Borel set P with $\nu^-(P) = 0$ and then, by inner regularity, compact $F \subseteq P$ with $\nu^+(F) > 0$. By heredity of E (proved below), applied to F and $\nu^+ \llcorner F$, there is compact zero-dimensional $Z \subseteq F$ with $\nu^+(Z) > 0$. Since $\nu^-(F) = 0$, one has $\nu(Z) > 0$.

For a nonzero complex measure ν , choose $\theta \in \mathbb{R}$ so that the real measure $\operatorname{Re}(e^{-i\theta}\nu)$ is nonzero. The real case gives Z with $\operatorname{Re}(e^{-i\theta}\nu(Z)) \neq 0$, hence $\nu(Z) \neq 0$. $\square$

4 The metric case and compact G_δ extraction

Definition 4.1. *A compact Hausdorff space K belongs to E_δ if every nonzero finite positive Radon measure on K charges a compact zero-dimensional G_δ subset of K .*

Exactly the supremum argument from Theorem 3.2 shows that $K \in E_\delta$ if and only if every positive μ can be approximated arbitrarily well by compact zero-dimensional G_δ subsets. Here one uses that a finite union of G_δ sets is G_δ and that restriction by a Borel set preserves Radon regularity.

Theorem 4.2 (Compact metrizable spaces). *Every compact metrizable space belongs to E_δ .*

Proof. Let K be compact metrizable, let μ be a finite positive Radon measure, and let $\eta > 0$. The one-point case is trivial. Fix a compatible metric d , let $D = \operatorname{diam} K > 0$, and choose a countable dense set $\{d_n : n \in \mathbb{N}\}$.

For a fixed center d_n , the spheres

$$S(d_n, r) = \{x : d(x, d_n) = r\}, \quad 0 < r < D,$$

are pairwise disjoint. Consequently, only countably many radii have positive μ -measure. Choose distinct radii $(r_{n,j})_{j \in \mathbb{N}}$ outside that exceptional set so that every tail $\{r_{n,j} : j \geq k\}$ is dense in $(0, D)$. Put

$$Y = K \setminus \bigcup_{n,j \in \mathbb{N}} S(d_n, r_{n,j}).$$

Then $\mu(K \setminus Y) = 0$. Every ball $B(d_n, r_{n,j}) \cap Y$ is clopen in Y . These sets form a base. Indeed, if $y \in O$ with O open, choose $\rho > 0$ with $B(y, \rho) \subseteq O$, then choose d_n with $d(y, d_n) < \rho/3$, and finally choose a tail radius $r_{n,j}$ in the nonempty interval $(d(y, d_n), \rho - d(y, d_n))$. Then $y \in B(d_n, r_{n,j}) \subseteq O$. Thus Y is zero-dimensional.

By inner regularity choose compact $Z \subseteq Y$ with $\mu(K \setminus Z) < \eta$. Then Z is zero-dimensional. Since K is metrizable, every compact subset is a G_δ subset of K . Hence $K \in E_\delta$. $\square$

Remark 4.3. *Zindulka proved the stronger metric-space statement that every finite measure on an arbitrary metric space has a zero-dimensional subspace of full measure [8]. The preceding proof is included because it directly repairs the local-base argument needed in the source paper and also produces compact approximants.*

5 Closure properties and the source-paper extensions

Theorem 5.1 (Countable closure). *The following assertions hold.*

- (1) $\mathbf{E}$ is hereditary for compact subspaces, closed under countable closed covers, and closed under countable products.
- (2) $\mathbf{E}_\delta$ is hereditary for compact subspaces and closed under countable products.
- (3) Both classes are closed under inverse limits indexed by a countable set (in particular, by a countable ordinal).

Proof. For heredity, let $F \subseteq K$ be compact and extend a measure ν on F to a measure on K supported by F . An extracting compactum Z may be replaced by $Z \cap F$. In the $\mathbf{E}_\delta$ case, $Z \cap F$ is G_δ in the subspace F .

If $K = \bigcup_n F_n$, where the F_n are compact and belong to $\mathbf{E}$, then a nonzero positive measure on K has $\mu(F_n) > 0$ for some n , and extraction inside F_n proves $K \in \mathbf{E}$.

For countable products, let $K = \prod_{n \in \mathbb{N}} K_n$, let μ be positive, and normalize to a probability measure. Write $\mu_n = (p_n)_* \mu$. By Theorem 3.2, choose compact zero-dimensional $A_n \subseteq K_n$ with

$$\mu_n(K_n \setminus A_n) < \eta 2^{-n-1}.$$

Then $A = \prod_n A_n$ is compact zero-dimensional and

$$\mu(K \setminus A) \leq \sum_n \mu_n(K_n \setminus A_n) < \eta.$$

If every $K_n \in \mathbf{E}_\delta$, choose the A_n to be G_δ . Then

$$A = \bigcap_n p_n^{-1}(A_n)$$

is G_δ in K .

Finally, an inverse limit of a countable system of compact Hausdorff spaces is a closed subspace of their countable product. Apply product closure and heredity. $\square$

Recall that an extension in [3, Definition 5.1] is the closure in $K \times [0, 1]$ of the graph of a function defined on a dense open subset of K .

Corollary 5.2 (All extensions preserve extraction). *Let L be any extension of a compact Hausdorff space K by a pairwise disjoint sequence of $[0, 1]$ -valued continuous functions, in the sense of [3, Definition 5.1]. Then*

$$K \in \mathbf{E} \implies L \in \mathbf{E}, \quad K \in \mathbf{E}_\delta \implies L \in \mathbf{E}_\delta.$$

No strongness assumption and no hypothesis on $K \setminus D((f_n))$ is needed.

Proof. By definition, L is compact and therefore closed in $K \times [0, 1]$. The interval belongs to $\mathbf{E}_\delta$ by Theorem 4.2. Apply Theorem 5.1 to the product and then to the compact subspace L . $\square$

Corollary 5.3 (Countable towers). *Start with a space in $\mathbf{E}$ (respectively, $\mathbf{E}_\delta$), take source-paper extensions at successor stages, and take inverse limits at limit stages. Every stage indexed by a countable ordinal remains in $\mathbf{E}$ (respectively, $\mathbf{E}_\delta$).*

Proof. Use transfinite induction, Corollary 5.2 at successor stages, and countable inverse-limit closure at limit stages. $\square$

Remark 5.4 (What this does and does not repair). *Corollary 5.3 removes the G_δ /countable-character obstruction identified in the partial packet: compact zero-dimensional G_δ sets persist through every countable stage. It does not make the source construction nonmetrizable for free. At the beginning of the proof of [3, Proposition 6.2], a bounded sequence of measures is passed to a weak-star convergent subsequence. That is a sequential compactness input supplied by metrizability, and later selection lemmas also use metric structure. Thus the present closure theorem is relevant but does not settle the ZFC “few operators” construction.*

6 A literature counterexample from a normal measure

A finite Radon measure μ on a compact space K is called *normal* (or hyperdiffuse) if $\mu(B) = 0$ for every Borel set B with empty interior. Plebanek constructed in ZFC a nondegenerate compact connected space carrying a normal probability measure [6, 7].

Proposition 6.1. *If a nondegenerate compact connected space K carries a normal probability measure, then $K \notin \mathbf{E}$.*

Proof. Let $F \subseteq K$ be compact and zero-dimensional. We claim that F has empty interior. Otherwise choose $x \in \text{int}_K F$ and a point $y \in K \setminus \{x\}$. Since F has a clopen base, there is a nonempty set C clopen in F such that

$$x \in C \subseteq (\text{int}_K F) \setminus \{y\}.$$

Because F is closed, C is closed in K ; because C is relatively open in F and lies in $\text{int}_K F$, it is also open in K . It is proper because $y \notin C$. This gives a nontrivial clopen subset of the connected space K , a contradiction. Hence every compact zero-dimensional F has empty interior, so normality gives $\mu(F) = 0$. The measure μ witnesses $K \notin \mathbf{E}$. $\square$

This is a counterexample to any conjecture that all compact Hausdorff spaces have the extraction property. It is not, by itself, a classification of the spaces that do.

7 Supports and product measures

For a positive Radon measure μ on a compact Hausdorff space K , define

$$\text{supp } \mu = K \setminus \bigcup \{U \subseteq K : U \text{ open and } \mu(U) = 0\}.$$

Lemma 7.1. *The support is compact and has full measure. Every nonempty relatively open subset of $\text{supp } \mu$ has positive μ -measure.*

Proof. Let U be the union of all open null sets. Every compact subset of U is covered by finitely many open null sets and is therefore null. Inner regularity on the open set U gives $\mu(U) = 0$. Thus $\text{supp } \mu = K \setminus U$ is compact and has full measure. The last assertion follows immediately from the definition, using that μ is concentrated on its support. $\square$

Definition 7.2. *A compact space K belongs to $\mathbf{S}_0$ if $\text{supp } \mu$ is zero-dimensional for every finite positive Radon measure μ on K .*

The class S_0 is much smaller than E . For instance, every zero-dimensional compactum belongs to S_0 , while a non-zero-dimensional compact metrizable space does not.

Lemma 7.3. *The class S_0 is hereditary for compact subspaces and closed under arbitrary products.*

Proof. Heredity follows by extending a measure from a compact subspace and observing that its support computed in the subspace is a closed subspace of the ambient support.

Let $K = \prod_{i \in I} K_i$ with every $K_i \in S_0$, and let μ be a positive measure on K . For the marginal $\mu_i = (p_i)_* \mu$, put $S_i = \text{supp } \mu_i$. If $x \in \text{supp } \mu$, then $x_i \in S_i$ for every i , since a null neighborhood of x_i would pull back to a null neighborhood of x . Hence

$$\text{supp } \mu \subseteq \prod_{i \in I} S_i.$$

Every S_i is zero-dimensional, an arbitrary product of zero-dimensional compacta is zero-dimensional, and closed subspaces of such compacta are zero-dimensional. Thus $\text{supp } \mu$ is zero-dimensional. $\square$

We shall use the canonical Radon product measure for an arbitrary family of compact spaces. If X_i carries a probability λ_i , integration on finite-coordinate continuous functions defines a positive norm-one functional on the algebra of continuous cylinder functions. Stone–Weierstrass makes this algebra uniformly dense in $C(\prod_i X_i)$, and the Riesz representation theorem gives a unique Radon probability $\bigotimes_i \lambda_i$ with the prescribed finite-dimensional marginals. If every λ_i is strictly positive, then the product measure is strictly positive, because every nonempty open set contains a nonempty basic cylinder.

The next lemma is the key point behind the exact product theorem.

Lemma 7.4 (A positive compact set contains a full tail fiber). *Let $X = \prod_{i \in I} X_i$, where every X_i is a nonempty compact Hausdorff space carrying a strictly positive Radon probability λ_i . Let $\lambda = \bigotimes_{i \in I} \lambda_i$. If $F \subseteq X$ is compact and $\lambda(F) > 0$, then there are a countable set $C \subseteq I$ and a point $a \in \prod_{i \in C} X_i$ such that*

$$\{a\} \times \prod_{i \in I \setminus C} X_i \subseteq F,$$

under the natural identification $X \cong X_C \times X_{I \setminus C}$.

Proof. Put $U = X \setminus F$. For every $n \geq 1$, choose compact $H_n \subseteq U$ such that

$$\lambda(H_n) > \lambda(U) - 2^{-n}.$$

Each H_n is covered by finitely many basic open cylinders contained in U . Let V be the union of all those cylinders. Then $V \subseteq U$, the set V depends on only countably many coordinates, and

$$\lambda(V) = \lambda(U).$$

Thus there are a countable $C \subseteq I$ and an open set $B \subseteq X_C$ such that $V = p_C^{-1}(B)$. Put $A = X_C \setminus B$ and

$$P = p_C^{-1}(A) = A \times X_{I \setminus C}.$$

We have $F \subseteq P$ and $\lambda(P \setminus F) = 0$.

Write $T = I \setminus C$, $\lambda_C = \bigotimes_{i \in C} \lambda_i$, and $\lambda_T = \bigotimes_{i \in T} \lambda_i$. By uniqueness of the product measure, $\lambda = \lambda_C \otimes \lambda_T$. Let $\nu = \lambda_C \perp A$. Since $\lambda(P) \geq \lambda(F) > 0$, the measure ν is nonzero; choose $a \in \text{supp } \nu$. The measure λ_T is strictly positive, so

$$\text{supp}(\nu \otimes \lambda_T) = \text{supp } \nu \times X_T.$$

The relatively open set $P \setminus F$ is null for $\nu \otimes \lambda_T$, and hence misses its support. Therefore

$$\{a\} \times X_T \subseteq \text{supp } \nu \times X_T \subseteq F.$$

□

8 An exact theorem for arbitrary products

Theorem 8.1 (Product characterization). *Let $(K_i)_{i \in I}$ be nonempty compact Hausdorff spaces and put $K = \prod_{i \in I} K_i$. Then $K \in \mathbf{E}$ if and only if both of the following conditions hold:*

- (a) $K_i \in \mathbf{E}$ for every $i \in I$;
- (b) the set

$$J = \{i \in I : K_i \notin \mathbf{S}_0\}$$

is countable.

Proof. Assume (a) and (b), and let μ be a nonzero positive measure on K . For every i , let $\mu_i = (p_i)_* \mu$ and $S_i = \text{supp } \mu_i$. By Lemma 7.1, $\text{supp } \mu$ has full measure, and

$$\text{supp } \mu \subseteq P := \prod_{i \in I} S_i.$$

For $i \notin J$, the compactum S_i is zero-dimensional by the definition of $\mathbf{S}_0$. For $i \in J$, the compactum S_i belongs to $\mathbf{E}$ by heredity from K_i . Since J is countable, $\prod_{i \in J} S_i \in \mathbf{E}$ by Theorem 5.1; the product $\prod_{i \notin J} S_i$ is zero-dimensional, hence belongs to $\mathbf{E}$. A two-factor application of product closure gives $P \in \mathbf{E}$. Restricting μ to the full-measure compact set P now yields the required zero-dimensional compact subset.

Conversely, suppose $K \in \mathbf{E}$. Each factor K_i is homeomorphic to a compact subspace of K after fixing base points in the other coordinates, so heredity gives (a). If (b) failed, there would be an uncountable $J \subseteq I$ such that $K_i \notin \mathbf{S}_0$ for every $i \in J$. Choose a probability μ_i on K_i whose support S_i is not zero-dimensional, and regard μ_i as a strictly positive probability on S_i . On

$$X = \prod_{i \in J} S_i$$

form the product probability $\lambda = \bigotimes_{i \in J} \mu_i$.

Let $F \subseteq X$ be compact with $\lambda(F) > 0$. Lemma 7.4 gives a countable $C \subseteq J$ and a full tail fiber contained in F . Choose $i \in J \setminus C$. By fixing all tail coordinates except the i th, that fiber contains a closed copy of S_i . Since S_i is not zero-dimensional, neither is F . Thus λ vanishes on every compact zero-dimensional subset of X , so $X \notin \mathbf{E}$. But X is a compact subspace of K , contradicting heredity. Hence J must be countable. □

Remark 8.2. *The proof uses only standard facts about Radon product measures, but I did not locate Theorem 8.1 in this exact form in the searched literature. It should therefore be treated as a proved result of this note, not as a literature attribution or a claim of publication-level novelty.*

9 Consequences and sharp counterexamples

Proposition 9.1. *For a compact metrizable space M , one has $M \in \mathbf{S}_0$ if and only if M is zero-dimensional.*

Proof. The forward implication is the only nontrivial one. Every compact metrizable space has a countable dense set (x_n) . The probability

$$\mu = \sum_{n=1}^{\infty} 2^{-n} \delta_{x_n}$$

is strictly positive and has support M . If $M \in \mathbf{S}_0$, this support must be zero-dimensional. $\square$

Corollary 9.2 (Products of metrizable compacta). *For nonempty compact metrizable spaces $(M_i)_{i \in I}$,*

$$\prod_{i \in I} M_i \in \mathbf{E} \iff \{i \in I : \dim M_i > 0\} \text{ is countable.}$$

In particular,

$$[0, 1]^I \in \mathbf{E} \iff I \text{ is countable.}$$

Proof. Every compact metrizable factor belongs to $\mathbf{E}$ by Theorem 4.2, and Proposition 9.1 identifies exactly which factors belong to $\mathbf{S}_0$. Apply Theorem 8.1. $\square$

Corollary 9.3 (A concrete witnessing measure). *If I is uncountable, the canonical product Lebesgue probability on $[0, 1]^I$ assigns measure zero to every compact zero-dimensional subset.*

Proof. A positive compact subset would contain, by Lemma 7.4, a full tail fiber outside countably many coordinates. That fiber contains a closed copy of $[0, 1]$, so the compact set cannot be zero-dimensional. $\square$

Corollary 9.4 (Sharpness of countable inverse-limit closure). *The cube $[0, 1]^{\omega_1}$ is compact, connected, and separable, and it is the inverse limit of the compact metrizable cubes $[0, 1]^\alpha$ for $\alpha < \omega_1$. Nevertheless, $[0, 1]^{\omega_1} \notin \mathbf{E}$.*

Proof. The inverse-limit statement is given by the coordinate projections, and each $\alpha < \omega_1$ is countable. Connectedness is preserved by products. Separability follows from the Hewitt–Marczewski–Pondiczery theorem, since $\omega_1 \leq 2^{\aleph_0}$ and every factor is separable [5]. Failure of $\mathbf{E}$ follows from Corollary 9.3. $\square$

Thus Theorem 5.1 is sharp with respect to the length of the inverse system: “countable” cannot be replaced by ω_1 , even for a separable connected limit whose every proper stage is metrizable and belongs to $\mathbf{E}_\delta$.

Corollary 9.5 (Continuous images). *The class $\mathbf{E}$ is not preserved by continuous images.*

Proof. Let I be uncountable and choose a continuous surjection from the Cantor set $2^{\mathbb{N}}$ onto $[0, 1]$. Its I -fold product maps the zero-dimensional compactum $(2^{\mathbb{N}})^I$ continuously onto $[0, 1]^I$, which fails $\mathbf{E}$ by Corollary 9.2. $\square$

10 Conclusion

The partial automatic solution was fundamentally on the right track, but its main hypothesis was misclassified as stronger than the original property and its extension theorem carried unnecessary assumptions. The corrected picture is:

- positive extraction, arbitrarily good compact zero-dimensional approximation, and countable decomposition into measures with zero-dimensional supports are equivalent;
- both ordinary extraction and compact G_δ extraction survive all countable products, countable inverse limits, and all countable towers of the source-paper extension operation;
- there are ZFC failures, including Plebanek's connected normal-measure compactum and the elementary separable connected cube $[0, 1]^{\omega_1}$;
- arbitrary products admit the exact characterization in Theorem 8.1.

These results substantially delimit the desired class but do not classify all compact Hausdorff spaces. In particular, the intrinsic structure of non-product compacta that fail extraction remains open in the generality of Problem 7.3.

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